

TAO LU

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👤 SUMMARY

Researcher in 3D vision and embodied AI, known for structured 3D Gaussian Splatting (Scaffold-GS and its follow-ups) and now building world action models and real-to-sim pipelines for robot learning.

- **Impact:** Scaffold-GS (CVPR 2024 Highlight) has 1,000+ citations and 1.3k GitHub stars; code released for 16 papers, with 4.8k+ GitHub stars in total.

📁 EXPERIENCE

Shanghai AI Laboratory, Researcher 2025 – present

- World action models: technical co-lead and corresponding author of InternW0- Δ (20K+ hours of open data), and Skel-WAM for hand-skeleton-conditioned human-to-robot transfer.
- Robot learning from videos: real-to-sim-to-real skills (Real2Gym, corresponding author) and world-space hand motion from egocentric video (InfiniHand).
- 3D reconstruction for embodied AI: streaming reconstruction (PLANING, NeurIPS 2026; M³; AquaFlow), world-consistent view synthesis (GeoVerse) and editable indoor scenes (EAG-PT, SIGGRAPH 2026).

Brown University, Postdoctoral Researcher 2024 – 2025

- Fast 4K Gaussian fitting (Turbo-GS, CVPR 2026 Highlight, co-first author), 4D volumetric video (PackUV, CVPR 2026) and visuotactile data for deformable world models (Deform360, ECCV 2026).

Shanghai AI Laboratory, Researcher 2023 – 2024

- Introduced Scaffold-GS, anchor-based neural Gaussians (CVPR 2024 Highlight), and extended it to city-scale level-of-detail rendering (Octree-GS, TPAMI 2025) and surface reconstruction (GSDF, NeurIPS 2024).

🎓 EDUCATION

Nanjing University, Ph.D. in Computer Science (point cloud and LiDAR perception) 2018 – 2023

Nanjing University, M.S. in Computer Science 2016 – 2018

Southwest Jiaotong University, B.E. in Automation 2012 – 2016

★ SELECTED PUBLICATIONS

* equal contribution † corresponding author

- **Real2Gym: Building Gyms from Videos, Bringing Skills to Robots**
arXiv 2026 · K. Ren*, Y. Xu*, K. Song, L. Xu, B. Dai, M. Yu†, **Tao Lu†**
- **InternW0- Δ : A World Action Model Bridging Predictive Dynamics and Actions with 20K+ Hours of Open Data**
arXiv 2026 · X. Miao, Z. Li, B. Fang, ..., J. Dong†, H. Guo†, **Tao Lu†**, M. Yu†, ..., W. Zhang†, C. Shen† (48 authors)
- **Turbo-GS: Accelerating 3D Gaussian Fitting for High-Quality Radiance Fields**
CVPR 2026, Highlight · A. Dhiman*, **Tao Lu***, R. Srinath*, E. Arslan, A. Xing, Y. Xiangli, R. V. Babu, S. Sridhar
- **Octree-GS: Towards Consistent Real-time Rendering with LOD-Structured 3D Gaussians**
TPAMI 2025 · K. Ren*, L. Jiang*, **Tao Lu**, M. Yu, L. Xu, Z. Ni, B. Dai†
- **GSDF: 3DGS Meets SDF for Improved Rendering and Reconstruction**
NeurIPS 2024 · M. Yu*, **Tao Lu***, L. Xu, L. Jiang, Y. Xiangli†, B. Dai
- **Scaffold-GS: Structured 3D Gaussians for View-Adaptive Rendering**
CVPR 2024, Highlight · **Tao Lu***, M. Yu*, L. Xu, Y. Xiangli, L. Wang, D. Lin, B. Dai†
- **LinK: Linear Kernel for LiDAR-based 3D Perception**
CVPR 2023 · **Tao Lu**, X. Ding, H. Liu, G. Wu, L. Wang†

⚙️ SERVICE & OPEN SOURCE

- **Reviewer:** CVPR, ICCV, ECCV, NeurIPS, SIGGRAPH, SIGGRAPH Asia, AAAI, IROS, TPAMI, IJCV.
- **Open source:** [Scaffold-GS](#) (1.3k stars), [AnySplat](#) (950), [Octree-GS](#) (860), [SparseBEV](#) (470), [GSDF](#) (470).

Conference papers

- [C1] **Turbo-GS: Accelerating 3D Gaussian Fitting for High-Quality Radiance Fields**
CVPR 2026, Highlight · A. Dhiman^{*}, **Tao Lu^{*}**, R. Srinath^{*}, E. Arslan, A. Xing, Y. Xiangli, R. V. Babu, S. Sridhar
- [C2] **PackUV: Packed Gaussian UV Maps for 4D Volumetric Video**
CVPR 2026 · A. Rai, A. Xing, A. Agarwal, X. Cong, Z. Li, **Tao Lu**, A. Prakash, S. Sridhar
- [C3] **EAG-PT: Emission-Aware Gaussians and Path Tracing for Diffuse Indoor Scene Reconstruction and Editing**
SIGGRAPH 2026 · X. Yang, M. Yu, C. Jiang, K. Ren, **Tao Lu**, J. Pang, D. Lin, B. Dai, L. Xu
- [C4] **TrajVG: 3D Trajectory-Coupled Visual Geometry Learning**
SIGGRAPH 2026 · X. Miao, W. Zhao, **Tao Lu**, L. Xu, M. Yu, Y. Long, J. Pang, J. Dong
- [C5] **SynthVerse: A Large-Scale Diverse Synthetic Dataset for Point Tracking**
SIGGRAPH 2026 · W. Zhao, H. Xu, X. Miao, Q. Zhao, R. Zhang, K. Huang, N. Gao, P. Cao, M. Sun, M. Yu, **Tao Lu**, L. Xu, J. Dong, J. Pang
- [C6] **Deform360: A Massive Multi-view Visuotactile Dataset for Deformable World Models**
ECCV 2026 · H. Li, W. Fu, X. Cong, Z. Li, B. Huang, H. Jiang, X. He, Y. Liang, R. Fu, **Tao Lu**, S. Sridhar, K. A. Smith, G. Konidaris, Y. Li
- [C7] **PLANING: A Loosely Coupled Triangle-Gaussian Framework for Streaming 3D Reconstruction**
NeurIPS 2026 · C. Jiang^{*}, K. Ren^{*}, X. Li, K. Song, G. Li, L. Xu, **Tao Lu**, J. Dong, Y. Zhang[†], Y. Feng, B. Dai, M. Yu[†]
- [C8] **AnySplat: Feed-forward 3D Gaussian Splatting from Unconstrained Views**
SIGGRAPH Asia 2025 (ACM TOG) · L. Jiang^{*}, Y. Mao^{*}, L. Xu, **Tao Lu**, K. Ren, Y. Jin, X. Xu, M. Yu, J. Pang, F. Zhao, D. Lin, B. Dai
- [C9] **GaRe: Relightable 3D Gaussian Splatting for Outdoor Scenes from Unconstrained Photo Collections**
ICCV 2025 · H. Bai, J. Zhu, S. Jiang, W. Huang, **Tao Lu**, Y. Li, J. Guo, R. Fu, Y. Guo, L. Chen
- [C10] **Horizon-GS: Unified 3D Gaussian Splatting for Large-Scale Aerial-to-Ground Scenes**
CVPR 2025 · L. Jiang^{*}, K. Ren^{*}, M. Yu, L. Xu, J. Dong, **Tao Lu**, F. Zhao, D. Lin, B. Dai
- [C11] **GSDF: 3DGS Meets SDF for Improved Rendering and Reconstruction**
NeurIPS 2024 · M. Yu^{*}, **Tao Lu^{*}**, L. Xu, L. Jiang, Y. Xiangli[†], B. Dai
- [C12] **Scaffold-GS: Structured 3D Gaussians for View-Adaptive Rendering**
CVPR 2024, Highlight · **Tao Lu^{*}**, M. Yu^{*}, L. Xu, Y. Xiangli, L. Wang, D. Lin, B. Dai[†]
- [C13] **SparseBEV: High-Performance Sparse 3D Object Detection from Multi-Camera Videos**
ICCV 2023 · H. Liu, Y. Teng, **Tao Lu**, H. Wang, L. Wang[†]
- [C14] **LinK: Linear Kernel for LiDAR-based 3D Perception**
CVPR 2023 · **Tao Lu**, X. Ding, H. Liu, G. Wu, L. Wang[†]
- [C15] **CamLiFlow: Bidirectional Camera-LiDAR Fusion for Joint Optical Flow and Scene Flow Estimation**
CVPR 2022, Oral · H. Liu, **Tao Lu**, Y. Xu, J. Liu, W. Li, L. Chen
- [C16] **CGA-Net: Category Guided Aggregation for Point Cloud Semantic Segmentation**
CVPR 2021 · **Tao Lu**, L. Wang[†], G. Wu

Journal papers

- [J1] **Octree-GS: Towards Consistent Real-time Rendering with LOD-Structured 3D Gaussians**
TPAMI 2025 · K. Ren^{*}, L. Jiang^{*}, **Tao Lu**, M. Yu, L. Xu, Z. Ni, B. Dai[†]
- [J2] **MixRF: Universal Mixed Radiance Fields with Points and Rays Aggregation**
TVCG 2025 · H. Bai, **Tao Lu**, J. Zhu, W. Huang, C. Gou, J. Guo, L. Chen, Y. Guo
- [J3] **Learning Optical Flow and Scene Flow with Bidirectional Camera-LiDAR Fusion**
TPAMI 2023 · H. Liu, **Tao Lu**, Y. Xu, J. Liu, L. Wang[†]
- [J4] **APP-Net: Auxiliary-Point-Based Push and Pull Operations for Efficient Point Cloud Recognition**
TIP 2023 · **Tao Lu**, C. Liu, Y. Chen, G. Wu, L. Wang[†]

Preprints

- [P1] **Real2Gym: Building Gyms from Videos, Bringing Skills to Robots**
arXiv 2026 · K. Ren^{*}, Y. Xu^{*}, K. Song, L. Xu, B. Dai, M. Yu[†], **Tao Lu[†]**
- [P2] **InternW0-Δ: A World Action Model Bridging Predictive Dynamics and Actions with 20K+ Hours of Open Data**
arXiv 2026 · X. Miao, Z. Li, B. Fang, ..., J. Dong[†], H. Guo[†], **Tao Lu[†]**, M. Yu[†], ..., W. Zhang[†], C. Shen[†] (48 authors)
- [P3] **Skel-WAM: A Hand-Skeleton-Conditioned World Action Model for Human-to-Robot Manipulation Transfer**
arXiv 2026 · Z. Cai, Y. Li, Y. Wang, X. Zhan, Y. Yang, H. Ma, K. Li, **Tao Lu**, J. Pang, L. Xu, D. Lin
- [P4] **InfiniHand: Streaming World-Space Hand Motion Estimation from Egocentric Video**
arXiv 2026 · K. Ren, K. Song, W. Zhao, Y. Wang, Y. Liu, B. Dai, H. Guo, C. Shen, M. Yu, **Tao Lu**, J. Dong
- [P5] **GeoVerse: World-Consistent Novel View Synthesis in Geometric Latent Space**
arXiv 2026 · K. Ren, **Tao Lu**, L. Xu, C. Jiang, M. Huang, C. Shen, M. Yu[†], B. Dai[†]

[P6] **Benchmarking and Enhancing Skill-Level Memory for Partially Observable Robotic Manipulation**

arXiv 2026 · Y. Shi, J. Yang, X. Yang, S. Zhang, Y. Zhu, **Tao Lu**, L. Wang

[P7] **AquaFlow: A Monocular Gaussian Splatting SLAM for Underwater Streaming Reconstruction**

arXiv 2026 · Y. Xu, K. Ren, W. Guo, C. Jiang, **Tao Lu**, L. Xu, M. Yu

[P8] **M³: Dense Matching Meets Multi-View Foundation Models for Monocular Gaussian Splatting SLAM**

arXiv 2026 · K. Ren, G. Li, C. Jiang, Y. Xu, **Tao Lu**, L. Xu, J. Dong, J. Pang, M. Yu, B. Dai

Workshop papers

[W1] **Proc-GS: Procedural Building Generation for City Assembly with 3D Gaussians**

CVPR 2025 Workshop · Y. Li, X. Ran, L. Xu, **Tao Lu**, M. Yu, Z. Wang, Y. Xiangli, D. Lin, B. Dai

[W2] **Art3D: Training-Free 3D Generation from Flat-Colored Illustration**

CVPR 2025 Workshop · X. Cong^{*}, J. Shen^{*}, Z. Li, R. Fu, **Tao Lu**, S. Sridhar